



ARM

IoT Project

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BATCH : S4 MCA
ROLL NO : 20

PROJECT SCOPE

- 1) The project focuses on developing a robotic arm that can interpret human hand gestures for controlling motions of the robotic arm.
 - 2) It has various real-world applications, such as industrial automation, assistive robotics, and educational demonstrations.
 - 3) Combines servo motors, Arduino, and Python programming to seamlessly translate gestures into mechanical actions with accuracy and efficiency.
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OBJECTIVES

- 1) Design and build a robotic arm controlled by hand gestures using computer vision and machine learning techniques.
- 2) To implement real-time hand tracking to detect and interpret gestures using OpenCV and Mediapipe.
- 3) To enable communication between the hand gestures and the robotic arm using PySerial.

NEED FOR THE PROJECT

- 1) Make robotic arm technology more accessible for scientists, educators, and professionals in various fields.
 - 2) Provide a user-friendly robotic arm for controlling using hand gestures.
 - 3) Enhance precision and flexibility in robotic tasks by offering real-time, accurate gesture tracking and arm movement control.
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COMPONENTS USED

1.HARDWARE

- 1) Arduino Uno R3
 - 2) MG90S Servo Motors
 - 3) Servo Motor Driver
 - 4) 5V Voltage Regulator
 - 5) Web Camera
 - 6) Cables and Connectors
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2. SOFTWARE

1) Python with Libraries:

- OpenCV and MediaPipe for gesture detection, and PySerial for serial communication.

2) Arduino IDE:

- For writing and uploading the control code to the Arduino Uno.



LITERATURE REVIEW

SI. NO.	Paper t itle	Authors	Publication Year	Journal/Con ference	Methodology	Models	Accuracy
1.	Gesture- Controlled Robotic Arm Utilizing OpenCV	Jedidiah Paterson, Dr. Ahmed Aldabbagh	2021	European Journal of Science and Technology	Utilized OpenCV and Raspberry Pi. The robotic arm incorporated 3D printing and servo motors.	OpenCV's contour detection and convex hull algorithms for hand gesture recognitio n.	real-time response and effective gesture recognition

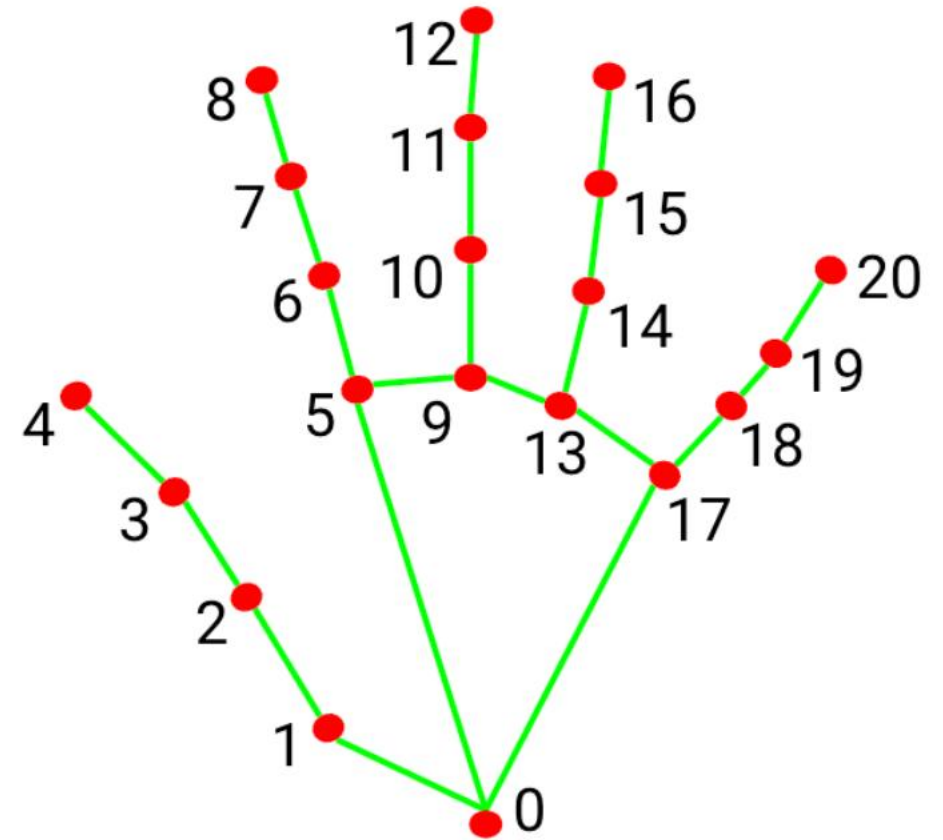
2.	Hand Gestures Robotic Control Based on Computer Vision	Marthed Wameed, Dr. Ahmed M. Alkamachi	2023	International Journal of Intelligent Systems and Applications in Engineering	MediaPipe for real-time hand gesture. used both raspberry and arduino.	MediaPipe framework with three models: palm detection, hand landmark recognition, and gesture classification.	100% accuracy for most gestures under ideal conditions (e.g : lighting at 100 lux, distance of 1m)
3.	A Study on Hand Motion Controlled Robotic Arm	Akarsh Kesharwani, Ayush P. Chaudhary, Bhanu Pratap Singh, Ved Kumar, Padmavathi M., Dr. Pavithra G., Dr. Sindhu Sree M., Dr. T. C. Manjunath	2023	Journal of Propulsion Technology	The study investigates hardware design, motion tracking algorithms, and user interface development to control a robotic arm using hand gestures.	Motion tracking and sensor fusion techniques for gesture recognition and robotic arm control.	High responsiveness and accuracy reported in real-time control scenarios

4.	Computer Vision-Based Hand Gesture Recognition for Human-Robot Interaction	Jing Qi, Li Ma, Zhenchao Cui, Yushu Yu	2024	Complex & Intelligent Systems	comprehensive review of hand gesture recognition using monocular and RGB-D cameras for human-robot interaction	dynamic time warping, hidden Markov models, machine learning classifiers (SVM), and deep learning methods	deep learning and RGB-D-based recognition systems showed high accuracy in specific applications
5.	Efficient Hand Gesture Recognition for Human-Robot Interaction	Marc Peral, Alberto Sanfeliu, Anaís Garrell	2022	IEEE Robotics and Automation Letters (RA-L)	Proposed a deep learning-based framework called EUREKA, which integrates MediaPipe for extracting 2D hand landmarks and a densely connected neural network for gesture classification.	MediaPipe for landmark extraction; DistTime feature representation and a Multi-Layer Perceptron with 128-64 hidden layer architecture for classification.	Achieved 87.5% accuracy

DATASET AND DATA PREPROCESSING

MediaPipe:

- Use MediaPipe BlazePalm (hand tracking) model.
- These models are often implemented using tensorflow Lite(wide range of devices).
- Once a hand is detected, a hand landmark model estimates 20 key points in 3D space.
- Works in real-time with OpenCV.



- Key landmarks is extracted from 20 hand landmark.
 - Wrist: landmarks(0)
 - Thumb: landmarks(1)–(4)
 - Index Finger: landmarks(5)–(8)
 - Middle Finger: landmarks(9)–(12)
 - Ring Finger: landmarks(13)–(16)
 - Little Finger: landmarks(17)–(20)
- Use dot product formula to calculate angle between the wrist, MCP joint, and PIP joint for each finger

$$\theta = \cos^{-1} \left(\frac{(A - B) \cdot (C - B)}{|A - B||C - B|} \right)$$

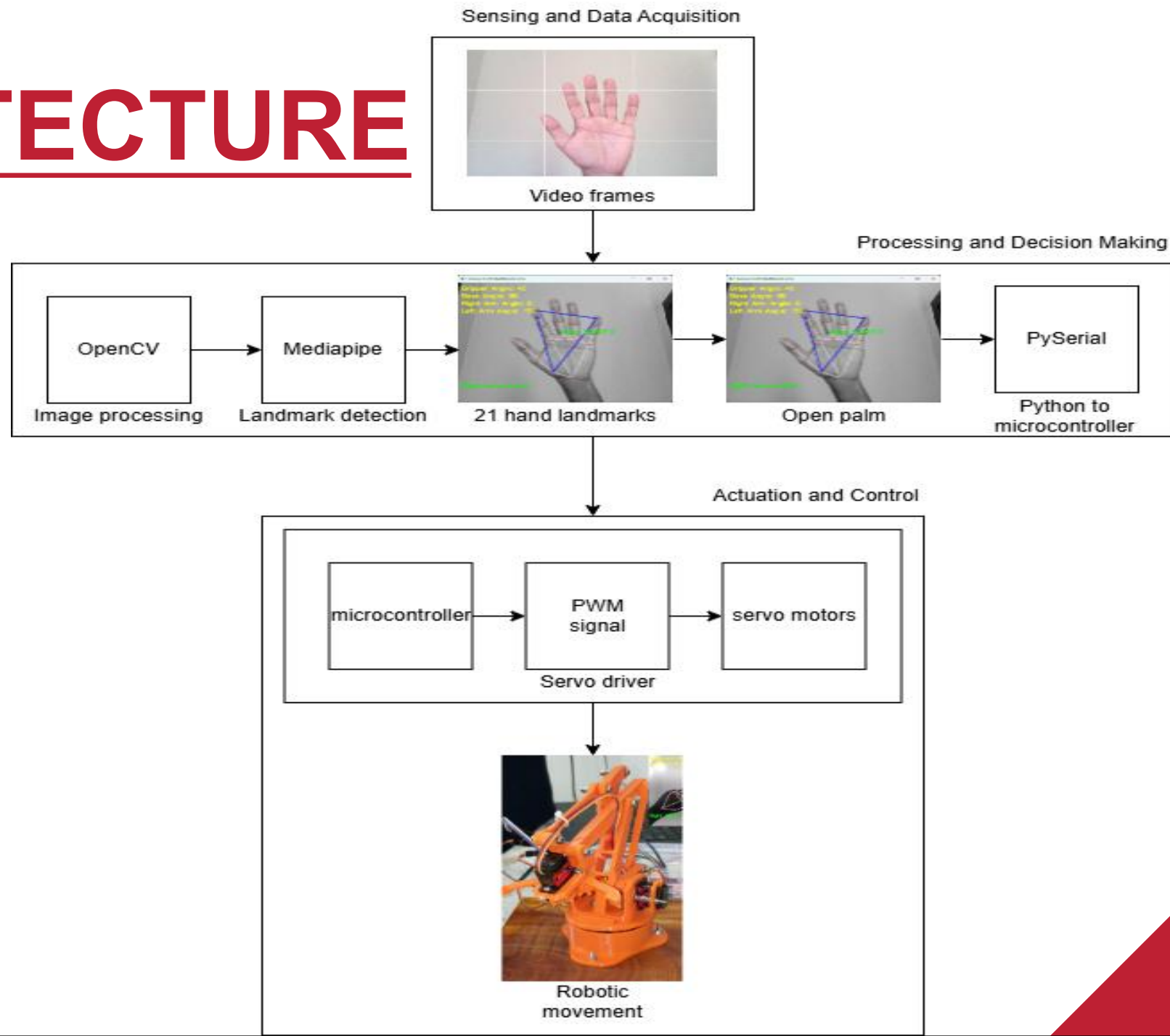
A=Base joint(MCP)

B=Middle joint(PIP)

C=Tip joint(DIP)

- Normalize the calculated angles to match the servo range 0° to 180° .
- Then convert to Servo Commands.

ARCHITECTURE



GESTURES

RIGHT ARM

- Arm move up - front movement
- Arm move down - back movement
- Arm move left - rotate left
- Arm move right - rotate right
- Palm open - to open claw
- Palm close - to close claw

LEFT ARM

- Index finger - left servo activate
- Index and middle finger - right servo activate





Shaper Angle: 40
Base Angle: 90
Right Arm Angle: 0
Left Arm Angle: 180

TOOLS

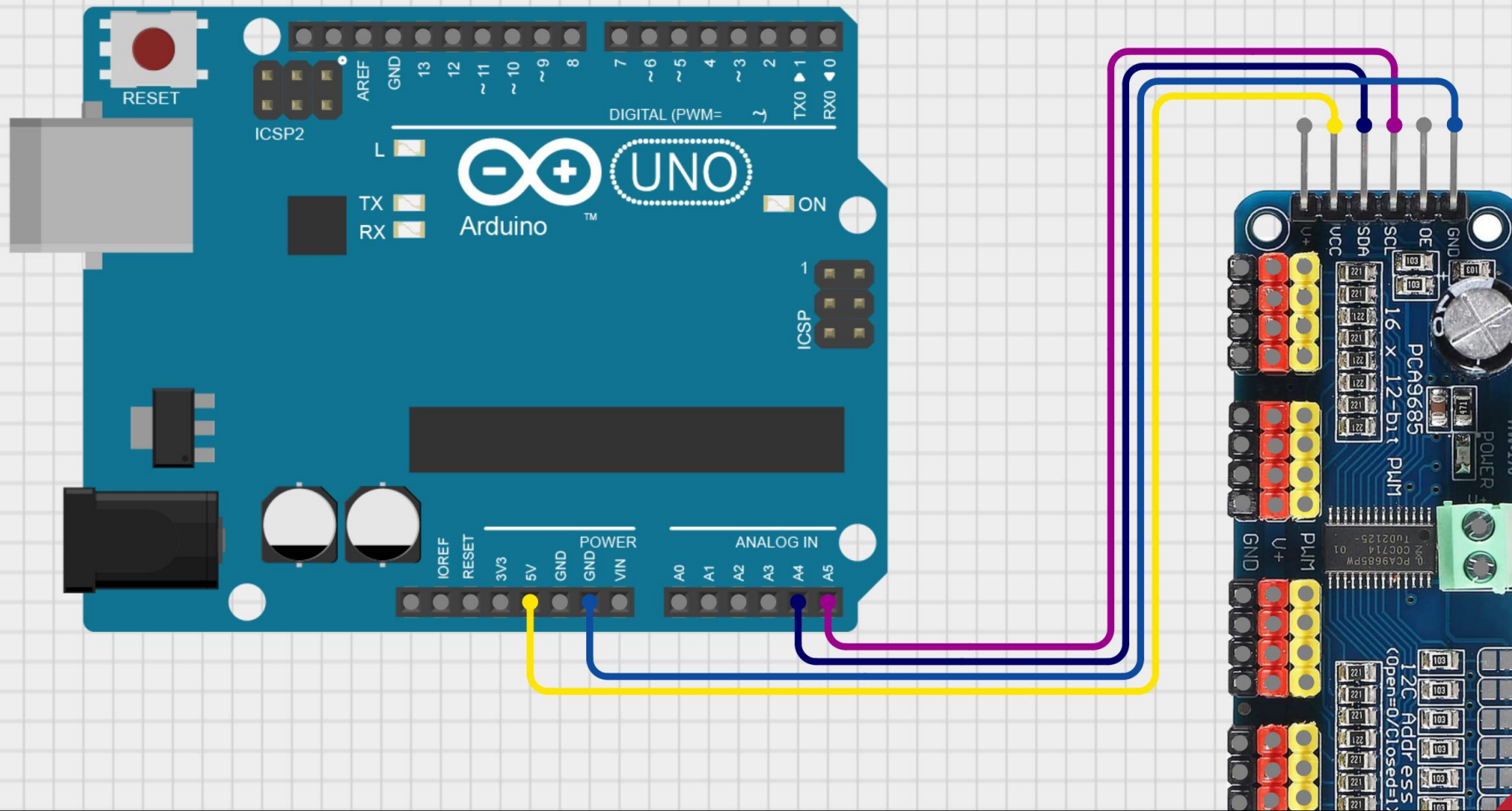
- **Fusion 360** : Design the parts required for the robotic arm.
- **Arduino IDE** : used to write and upload code to Arduino boards.



CIRCUIT DIAGRAM

ARDUINO - DRIVER

Cirkit Designer



Arduino - Driver

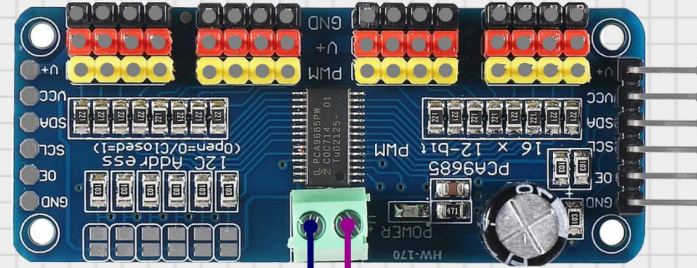
Gnd - Gnd

5v - Vcc

A5 - SCL

A4 - SDA

DRIVER - VOLTAGE REGULATOR

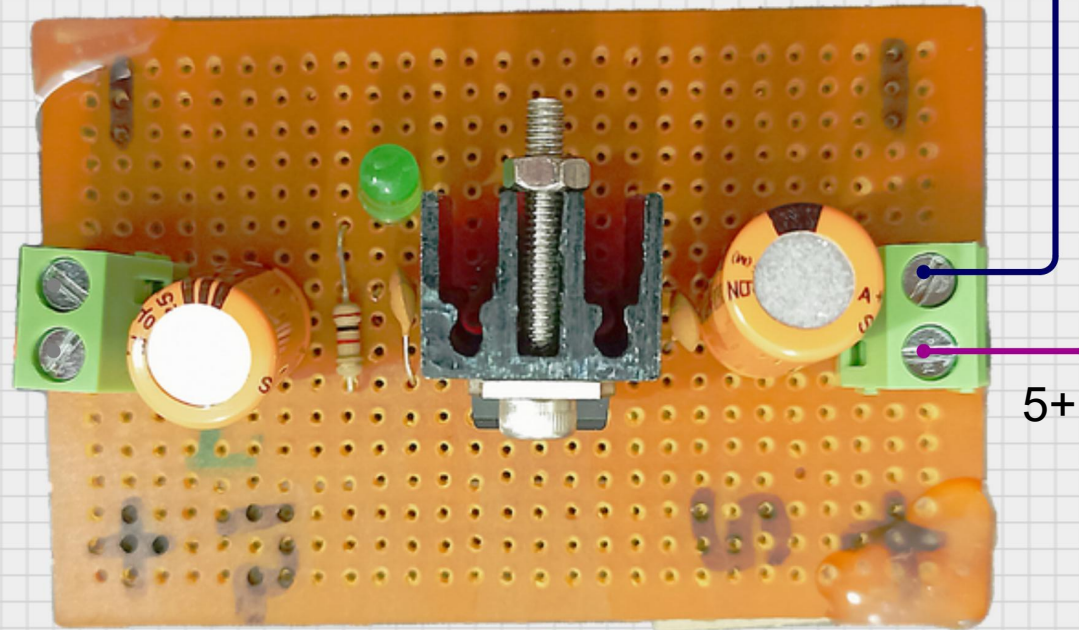


Vin+

Regulator - Driver

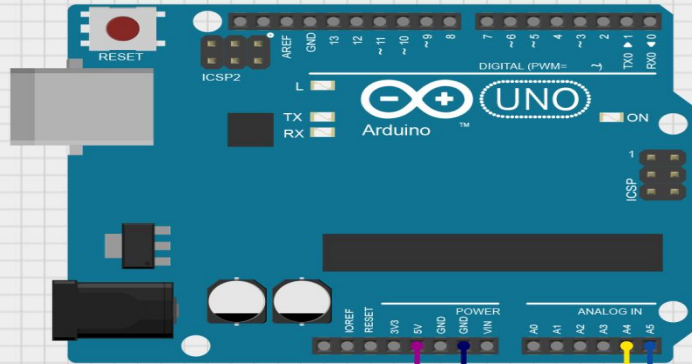
+5 - Vin+

Gnd - Gnd



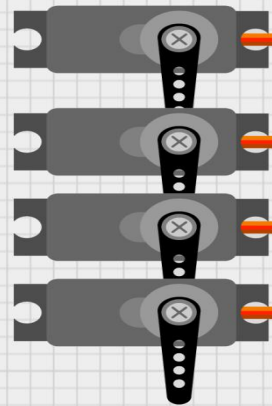
5+

Arduino UNO



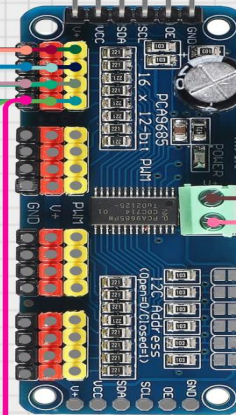
UNO - Driver

Gnd - Gnd
5v - Vcc
A5 - SCL
A4 - SDA

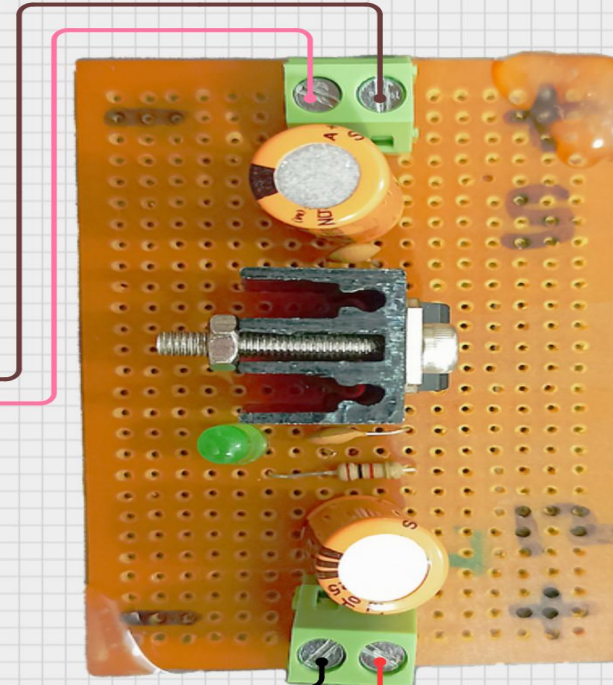


4 Servo motor

SERVO
DRIVER



12v to 5v Voltage Regulator



Regulator - Driver

+5 - Vin+
Gnd - Gnd

12v DC Adapter



ARM MODEL



References

- [1] M. Peral, A. Sanfeliu, and A. Garrell, "Efficient Hand Gesture Recognition for Human-Robot Interaction," IEEE Robotics and Automation Letters, vol. 7, no. 4, pp. 10272–10278, Oct. 2022, Link: <https://ieeexplore.ieee.org/document/9837407>.
- [2] J. Paterson and A. Aldabbagh, "Gesture-Controlled Robotic Arm Utilizing OpenCV," in 2021 International Conference on Harnessing Emerging Research for Opportunity (HORA), 2021, Link: <https://www.researchgate.net/publication/352903131>.
- [3] M. Wameed and A. M. Alkamachi, "Hand Gestures Robotic Control Based on Computer Vision," International Journal of Intelligent Systems and Applications in Engineering, vol. 11, no. 2, pp. 1013–1021, 2023. [Online]. Available: <https://www.researchgate.net/publication/381110677>.
- [4] J. Qi, L. Ma, Z. Cui, and Y. Yu, "Computer vision-based hand gesture recognition for human-robot interaction: a review," Complex & Intelligent Systems, vol. 10, pp. 1581–1606, 2024, Link: <https://link.springer.com/article/10.1007/s40747-023-01173-6>.
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THANK YOU